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Invited review

Impulsivity traits and neurocognitive mechanisms conferring vulnerability to substance use disorders

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ABSTRACT

Impulsivity - the tendency to act without sufficient consideration of potential consequences in pursuit of short-term rewards - is a vulnerability marker for substance use disorders (SUD). Since impulsivity is a multifaceted construct, which encompasses trait-related characteristics and neurocognitive mechanisms, it is important to ascertain which of these aspects are significant contributors to SUD susceptibility. In this review, we discuss how different trait facets, cognitive processes and neuroimaging indices underpinning impulsivity contribute to the vulnerability to SUD. We reviewed studies that applied three different approaches that can shed light on the role of impulsivity as a precursor of substance use related problems (versus a consequence of drug effects): (1) longitudinal studies, (2) endophenotype studies including non-affected relatives of people with SUD, and (3) clinical reference groups-based comparisons, i.e., between substance use and behavioural addictive disorders. We found that, across different methodologies, the traits of non-planning impulsivity and affect-based impulsivity and the cognitive processes involved in reward-related valuation are consistent predictors of SUD vulnerability. These aspects are associated with the structure and function of the medial orbitofrontal-striatal system and hyperexcitability of dopamine receptors in this network. The field still needs more theory-driven, comprehensive studies that simultaneously assess the different aspects of impulsivity in relation to harmonised SUD-related outcomes. Furthermore, future studies should investigate the impact of impulsivity-related vulnerabilities on novel patterns of substance use such as new tobacco and cannabinoid products, and the moderating impact of changes in social norms and lifestyles on the link between impulsivity and SUD.

This article is part of the special issue on 'Vulnerabilities to Substance Abuse'.

1. Introduction

Impulsivity is a predisposition towards rapid, unplanned and/or reward-driven reactions to internal or external triggers without sufficient consideration of the consequences to oneself and others (Moeller et al., 2001). Impulsivity has been consistently associated with the vulnerability to develop substance use disorders (SUD) and behavioural addictions (Castellanos-Ryan and Conrod, 2020; Goudriaan, 2020; Kozak et al., 2019; Verdejo-Garcia et al., 2008), but there is ongoing debate on which specific aspects of such broad predisposition drive this relationship (Enkavi et al., 2019; Vassileva and Conrod, 2019). There are several aspects of mental function, including trait-related characteristics and cognitive processes, that may contribute to impulsive behaviours and SUD susceptibility (MacKillop et al., 2016; Sharma et al., 2014).

Different personality theories have proposed distinct sets of traits underlying impulsivity. The most widely used frameworks in SUD research are the Barratt's impulsivity theory and the Urgency/Premeditation/Perseverance/Sensation seeking (UPPS) model (Barratt, 1965; Whiteside and Lynam, 2001) (Table 1). The Barratt's theory separates attentional, motor and non-planning facets (Patton et al., 1995), whereas the UPPS-P model segregates sensation seeking (including novelty seeking), emotion-driven (positive and negative urgency), and cognitive/conscientiousness related (lack of premeditation and perseverance) facets (Cyders et al., 2007; Whiteside and Lynam, 2001). There is notable overlap between these frameworks; for instance, non-planning and lack of premeditation facets are both linked to the construct of conscientiousness within the "big five" model of personality (Costa Jr and McCrae, 2008; Whiteside and Lynam, 2001). The positive and negative urgency dimensions were original contributions of the UPPS-P

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Table 1
Summary of theories, measures and behavioural and neuroimaging indices covered in the review.

Aspect	Model/ Domains	Measure	Behavioural Indices	Brain regions
Trait	Barratt's theory: Non-planning, Motor, Attention	Barratt Impulsivity Scale	Non-planning	Medial OFC (right)
			Motor	Superior OFC (left)
			Attentional	Parietal Cortex
	UPPS model: Urgency, lack of Premeditation, Perseverance, Sensation seeking	UPPS-P	Negative Urgency	MFG, Insula
			Premeditation	IFG, Insula, Putamen
			Perseverance	ACC
			Sensation Seeking	MFG, SmG
			Positive Urgency	Somatosensory II
	Four-factor model of personality vulnerability	Substance Use Risk Profile Scale	Impulsivity	–
Cognitive	Motor	Stop Signal	Stop Signal Reaction Time	IFG (right), preSMA Subthalamic Nucleus
		Go/No-go	Commission Errors	MFG/IFG, preSMA, Parietal Cortex, Insula, Putamen
		Immediate Memory Task	Commission Errors, Ratio catch/target trials	ACC
		Stroop	Interference score/time	ACC
	Choice	Delay Discounting	K parameter	IFG (left), DLPFC
		Balloon Analogue Risk Task	#Pumps	Insula, Caudate
		Monetary Incentive Delay	Reaction Time/Brain activation	VMPFC, VS, Thalamus, Insula, Amygdala
	Reflection	Matching Familiar Figures Test	Pre-decision time/Errors	VS, Thalamus
		4 Choice Task	Premature Errors	VS, Subgenual-ACC
		Information Sampling Task	Probability of Correct Response	Dorsal-ACC, Precuneus
		Tower tasks	Planning time/Errors	ACC, SMA, Rostrolateral PFC, Parietal Cortex
	Comprehensive	Cognitive Impulsivity Suite	Commission Errors	MFG
			Drift rate	Superior Parietal
			Perseveration	VMPFC

Note. Abbreviations of brain regions: OFC: Orbitofrontal Cortex; MFG: Middle Frontal Gyrus; IFG: Inferior Frontal Gyrus; SmG: Supramarginal Gyrus; SMA: Supplementary Motor Area; ACC: Anterior Cingulate Cortex; DLPFC: Dorsolateral Prefrontal Cortex; VS: Ventral Striatum; VMPFC: Ventromedial Prefrontal Cortex.
Brain regions associated with trait measures based on structural neuroimaging studies (Besteher et al., 2019; Holmes et al., 2016; Matsuo et al., 2009; Moreno-Lopez et al., 2012a, 2012b; Wang et al., 2017). Brain regions associated with cognitive measures based on functional magnetic resonance imaging (fMRI) studies measuring activation during the tasks, or meta-analysis of these studies, when available (Banca et al., 2016; Korucuoglu et al., 2020; Morris et al., 2016; Newman et al., 2009; Oldham et al., 2018; Rae et al., 2014; Ramaekers et al., 2016; Simmonds et al., 2008; Smith et al., 2018; Wagner et al., 2006).

model, although based on the neuroticism trait of the "big five" model (Costa Jr and McCrae, 2008; Whiteside and Lynam, 2001). Neuroimaging studies have shown that the different traits proposed by the Barratt and UPPS-P theories have some unique and some common brain structural underpinnings. They encompass regions of the prefrontal and parietal cortices, insula and striatum, although the findings are not conclusive (Besteher et al., 2019; Holmes et al., 2016; Matsuo et al., 2009; Moreno-Lopez et al., 2012a, 2012b; Wang et al., 2017). As an alternative to the Barratt's and UPPS-P propositions, some authors have posited simplified two-factor models that distinguish reward sensitivity (i.e. the tendency to seek immediate gratification) from disinhibition (i.e. general difficulty to regulate reckless behaviour), subserved by subcortical and cortical regions respectively (Gullo et al., 2014; Reise et al., 2013).

In the field of neuropsychology and cognitive neuroscience, impulsivity equates to weakened inhibitory control, which reflects the (in)ability of top-down cortical systems to regulate prepotent motor or reward-driven responses (Verdejo-Garcia et al., 2008). Inhibitory control sits under the umbrella construct of executive functions, together with working memory (i.e., the ability to maintain and manipulate information online) and flexibility (i.e., the ability to shift mental states and behavioural patterns) (Miyake and Friedman, 2012; Miyake et al., 2000). Altogether, executive functions act as a collection of higher-order cognitive processes that orchestrate goal-orientated behaviours (Friedman and Miyake, 2017; Verdejo-Garcia and Bechara, 2010). In this context, impulsivity may stem from at least two separate cognitive processes: rapid-response or motor impulsivity, defined as a tendency towards immediate action that occurs without enough forethought or disregard of current demands, and choice impulsivity, consisting of preferential selection of smaller sooner rewards over larger later ones (Hamilton et al., 2015a, 2015b; MacKillop et al., 2016) (Table 1). There is preliminary evidence that these two processes have segregated neural correlates, namely, for impulsive action: the anterior cingulate cortex, pre-supplementary motor area and inferior frontal gyrus, and for impulsive choice: the middle frontal gyrus and frontal pole (Wang et al., 2016). However, there is also considerable neural overlap between impulsive action and choice within the prefrontal cortex, striatum and insula (Korucuoglu et al., 2020; Rae et al., 2014; Simmonds et al., 2008; Smith et al., 2018). Several other theoretical frameworks have emphasised the contribution of additional cognitive processes, such as attentional control, reflection impulsivity and decision-making (Sharma et al., 2014; Voon, 2014). Reflection and uncertainty and risk-based decision-making processes engage ventromedial prefrontal cortex, subgenual cingulate cortex and ventral striatal regions, which are partly dissociable from the neural correlates of motor and choice impulsivity (Banca et al., 2016; Haber and Knutson, 2010; Morris et al., 2016; Oldham et al., 2018; Ramaekers et al., 2016). The planning or pre-decisional process preceding actions or choices additionally engages dorsal and rostrolateral aspects of the prefrontal cortex and parietal regions (Newman et al., 2009; Wagner et al., 2006). Different forms of impulsivity have been linked to the frontostriatal dopaminergic system, and specifically to the availability of D2-type receptors in the striatum, amygdala, insula and medial prefrontal cortex (Jaworska et al., 2020; Leyton and Vezina, 2014; London, 2020).

The trait facets underlying impulsivity are typically measured with self-report surveys. The most commonly used self-reports are the Barratt impulsivity scale (BIS) and the UPPS-P scale (Patton et al., 1995; Reise et al., 2013; Whiteside and Lynam, 2003). The Substance Use Risk Profiling Scale (SURPS) is also frequently used in SUD research and includes an impulsivity scale based on the Eysenck's impulsiveness facet and correlated with neuroticism and conscientiousness factors (Woicik et al., 2009). Impulsive action and choice are measured with laboratory tasks such as the stop signal and delay discounting tasks respectively (Hamilton et al., 2015a, 2015b). Studies have used several other tasks to measure cognitive processes related to impulsivity, including attentional/interference control tasks such as the Stroop and the immediate memory tasks; reflection/planning tasks such as the matching familiar figures test and tower tests; and reward-related valuation/decision-making tasks such as the monetary incentive delay task (MID) (Mechelmans et al., 2017; Verdejo-Garcia et al., 2008). The correlation between trait and cognitive measures of impulsivity is consistently low (Cyders and Coskunpinar, 2011; Eisenberg et al., 2019; Sharma et al., 2013), which (notwithstanding the conceptual hurdle) suggests that both provide independent contributions to the relationship between impulsivity and SUD (Sharma et al., 2014; Vassileva and Conrod, 2019). These independent contributions and their relative strength are relatively underexplored in the context of SUD risk. Therefore, a key unresolved question is which trait and cognitive aspects better explain the link between impulsivity-related predispositions and SUD-related manifestations.

In this review, we discuss how different trait facets, cognitive processes and related neuroimaging markers underpinning impulsivity contribute to the vulnerability to substance use related problems, including proxies of problematic alcohol and drug use (e.g., binge and hazardous drinking, frequency and escalation of illicit drug use) and SUD. To achieve this aim, we reviewed studies that applied three different approaches that can shed light on the role of impulsivity as a precursor of substance use related problems (versus a consequence of drug effects): (1) longitudinal studies, (2) endophenotype studies, and (3) clinical reference groups based comparisons, i.e., between substance use and behavioural addictive disorders. Longitudinal studies can examine the relationship between impulsivity measures collected prior to onset of substance use and subsequent risk of developing SUD over time. An additional advantage of this approach is the potential to identify moderators and mediators of this relationship, such as background characteristics, substance use related expectancies and motives, and social settings and norms. Endophenotype studies can reveal which aspects of impulsivity manifest in non-affected relatives of people with SUD (i.e., offspring or siblings). Such aspects reflect shared vulnerability to SUD, which can be separated from SUD-specific consequences that manifest only in the affected relatives. Finally, comparisons between people with SUD and those with behavioural addictions provide a naturalistic model of shared vulnerabilities, reflected in common impulsive characteristics across both disorders. Conversely, differences between the groups would putatively reflect the specific consequences or the direct impacts of each disorder. Altogether, these three strands of evidence have the potential to yield new insights to qualify the link between trait and cognitive impulsivity related vulnerabilities and SUD. We were inclusive in considering the different trait facets included in the Barratt and UPPS-P models of impulsivity (Patton et al., 1995; Whiteside and Lynam, 2001), and the distinct cognitive processes included in current neurocognitive frameworks, such as attention, reflection, inhibition, and choices involving risk and reward (Sharma et al., 2014; Vassileva and Conrod, 2019). Since we used similar sources of evidence in a comprehensive review published in 2008 (Verdejo-Garcia et al., 2008), here we focused on studies published after 2008, and preferably within the last 10 years.

2. Longitudinal studies on the link between premorbid impulsivity and emergence of substance use problems

Longitudinal designs provided original and then consistent evidence that impulsivity, measured during childhood and early adolescence, is a predictor of subsequent alcohol and drug use, and development of SUD (Dawes et al., 1997; Nigg et al., 2006; Tarter et al., 2003). Early studies focused on high-risk cohorts, whereby participants had a family history of SUD and other externalising disorders, and used multi-informant measures of impulsivity (i.e., children, parents and teachers) and cognitive tasks. Their findings were pioneering in showing that impulsive characteristics, identified as early as by 10 years old, successfully predicted initial involvement with substance use and later emergence of SUD. More recent studies have qualified these findings by providing in-depth examination of the traits and cognitive processes that contribute to this relationship, as well as its moderators and mediators; engaging both at-risk and community-based samples; and incorporating novel neuroimaging techniques and prediction models such as machine learning algorithms.

The studies investigating trait facets have corroborated the longitudinal relationship between impulsive characteristics measured during childhood and early adolescence, and alcohol and substance use in late adolescence. In community samples, the five facets of the UPPS model are significant predictors of frequency of drinking and smoking and alcohol-related problems (Jones et al., 2020; Peterson and Smith, 2017). In high-risk samples such as children with ADHD, the dimensions of positive and negative urgency specifically mediate the relationship between ADHD and alcohol-related problems (Pedersen et al., 2016). When impulsivity levels are initially measured by late adolescence, positive urgency and SURPS-impulsivity (similar to negative urgency and conscientiousness) are the most significant predictors of subsequent alcohol-related problems (Kaiser et al., 2016; Mackinnon et al., 2014). Furthermore, individuals with increasing lack of premeditation across adolescence are at higher risk of drinking problems and illicit drug use (Martinez-Loredo et al., 2018).

Cognitive tasks, compared to trait measures, are less consistent predictors of subsequent SUD related problems (Fernandez-Artamendi et al., 2018; Jones et al., 2020). This could be due to reduced ability of cognitive tasks to capture variability in impulsive characteristics, especially in neurotypical populations, and/or the lack of similarity between the assessment approach of cognitive tasks, which measure indirect indices of latent mechanisms, and drug use surveys (Enkavi et al., 2019). The latter difference contrasts with the similar methodology of trait and drug use measures (both self-reports). It is also possible that the lower conceptual development of cognitive versus trait impulsivity is impacting on the predictive validity of cognitive measures (Eisenberg et al., 2019). There is preliminary evidence that some indices of motor impulsivity, such as behavioural performance on the stop signal task and reduced prefrontal cortex activation during the go/no-go task measured with fMRI predict substance use progression (Romer Thomsen et al., 2018). However, other well-validated behavioural indices of motor impulsivity, such as commission errors in the go/no-go task or inhibition scores on the Stroop and immediate memory tasks, do not predict alcohol/substance use levels or related problems during adolescence (Fernandez-Artamendi et al., 2018; Jones et al., 2020; Romer Thomsen et al., 2018). In a well-powered twin study assessing motor impulsivity, measures of impulsive action (e.g., stop signal), clustered as part of a latent executive function factor at age 17, were not longitudinal predictors of SUD during young adulthood, although they correlated with number of substances used and frequency of use within each measurement wave (Gustavson et al., 2017).

In regards to choice impulsivity, studies in cohorts of mid and late adolescents have failed to show significant associations between delay discounting and development of binge drinking or alcohol related problems (Bernhardt et al., 2017; Fernandez-Artamendi et al., 2018), despite using comprehensive assessments of discounting (Bernhardt et al., 2017). However, there is a significant relationship between delay discounting (i.e., choice impulsivity) and development of smoking problems (Audrain-McGovern et al., 2009) and more generally for externalising disorders (Acheson et al., 2011). The longitudinal association between delay discounting and SUD risk may be better explained in the context of other executive functions. In two studies that followed up a community sample of 387 adolescents cognitively assessed at 11–13 years old, steeper delay discounting (i.e., greater choice impulsivity) mediated the impact of reduced working memory on development of alcohol use problems by mid adolescence and SUD by late adolescence (Khurana et al., 2013, 2017). Delay discounting also interacts with noradrenaline levels - a key neurotransmitter for executive functions - in predicting drug use among young adults (Brody et al., 2014).

Recent studies within the general population have incorporated multi-site approaches and neuroimaging measures as part of multimodal assessment protocols including developmental and socio-demographic, personality, cognitive and neural indices. One of these studies is the IMAGEN consortium, which has a sample of ~700 adolescents. In one study of the consortium, they found that a combination of background characteristics, the personality facets of conscientiousness (i.e. non-planning in Barratt's and lack of premeditation in UPPS models [trait impulsivity measures]) and anxiety sensitivity, and structural (GM:WM ratio; parenchymal volume) and MID-evoked functional brain features (right precentral gyrus activation during reward outcome and inhibition failure and bilateral superior frontal gyrus during reward outcome) predicted future binge drinking (a proxy of AUD) (Whelan et al., 2014). Conversely, ventromedial prefrontal cortex activation during emotional reactivity and face recognition was a better predictor of current binge drinking, probably reflecting alcohol-related neuroadaptations (Whelan et al., 2014). Still within the IMAGEN cohort, a study in a subsample of 144 adolescents with high novelty seeking characteristics, corresponding to impulsivity traits, showed that lower conscientiousness, steeper delay discounting, and greater MID-evoked activity in midbrain, ventral striatum and dorsolateral prefrontal cortex at 14 years old predicted problematic drug use at 16 (Buchel et al., 2017). In a smaller-scale fMRI study, MID-evoked hyperactivation of the nucleus accumbens among children aged 8–12 years old was also associated with earlier exposure to drug use, i.e., before 18, in a small community-based sample enriched for family history of SUD (Cope et al., 2019). In another small-scale neuroimaging study of young stimulant users, structural brain alterations (i.e. smaller brain volumes) in the dorsal striatum, basolateral amygdala and medial prefrontal cortex were longitudinally associated with escalation of stimulant use and development of SUD (Becker et al., 2015).

During adolescence and young adulthood, brain maturation goes in tandem with increases in exploratory behaviour and progressive adaptation to social norms (Sparks, 2019). Longitudinal studies have illustrated how the link between impulsivity and vulnerability to SUD is moderated by social norms and mediated by positive drug expectancies. In a three-year follow up of early adolescents (12–15 years old), cognitive-reflection impulsivity performance in a planning task (Tower of London) interacted with social norms at predicting alcohol use during mid adolescence (Meisel et al., 2015). Specifically, the perception that drinking was acceptable among peers increased alcohol use particularly in adolescents with reduced reflection. However, the opposite effect was found for motor impulsivity, as the same perception of social norms increased alcohol use among adolescents with better stop signal performance, which might be able to adjust drinking bouts to match normative social expectations. These findings highlight that the link between different forms of cognitive impulsivity and risk of SUD during adolescence can be more multifaceted and complex than previously thought, and calls for more multidimensional assessments of cognitive impulsivity in longitudinal studies. A study in young adults, which measured four personality dimensions linked to SUD risk according to the SURPS model, found that impulsivity traits were the most consistent predictors of future drinking problems, relative to sensation seeking, anxiety sensitivity or hopelessness (Mackinnon et al., 2014). Furthermore, the longitudinal association between trait impulsivity and alcohol use was mediated by enhancement and social motives, reflecting participants' expectancies that alcohol consumption would be exciting and could help them being sociable, respectively. In addition to positive expectancies, negative emotions and stress are significant moderators of the relationship between impulsivity and SUD risk; for example, trait impulsivity interacts with cumulative stress at predicting problematic alcohol use in youths (Fox et al., 2010). These findings highlight the relationship between impulsivity and proximal factors of alcohol and drug use, such as substance-related expectancies, peer-related influences and stressful events.

Altogether, the findings of longitudinal studies suggest that traits associated with conscientiousness (i.e., lack of premeditation, non-planning impulsivity) and emotion-driven impulsivity (positive and negative urgency) are significant predictors of SUD vulnerability. The evidence on cognitive predictors of SUD risk is inconclusive, and more work is needed to harmonize measures and improve the reliability and predictive validity of cognitive tasks. Nonetheless, impulsive choice, in combination with other executive functions, and neuroimaging indices of prefrontal alterations linked to impulsive action and reward-related choices are longitudinally associated with substance-related problems.

3. Endophenotype studies measuring impulsive characteristics in at-risk groups

Impulsivity can be considered an endophenotype of SUD, i.e., an intermediate phenotype between genetic vulnerability and clinical manifestations, which is present in unaffected biological relatives of people with SUD (Verdejo-Garcia et al., 2008). Twin and sibling studies, in which one member of the pair suffers SUD and the other one is unaffected (i.e., indicator of shared underlying risk), can shed light on whether impulsivity features in both affected and unaffected siblings, suggesting a role in the vulnerability to SUD.

In a series of studies involving sibling pairs with the same biological parents, one affected with stimulant-SUD and one unaffected (i.e., shared vulnerability/risk), and healthy controls (i.e., no risk), Ersche and colleagues have pinpointed trait, cognitive and brain features associated with addiction vulnerability (i.e., shared between users and unaffected siblings). When examining trait characteristics, non-planning impulsivity was elevated in non-affected siblings compared to controls, although affected siblings also showed elevated levels relative to siblings (Ersche et al., 2010). Interestingly, affected siblings had elevated sensation seeking compared to non-affected siblings and controls, which suggests SUD-related impacts on this impulsivity trait (Ersche et al., 2010). In a later study that included trait measures of impulsivity and emotional sensitivity and cognitive measures of inhibitory control (motor impulsivity) and executive functions (working memory and planning), results showed that stop signal performance was poorer in both affected and unaffected siblings compared to controls (Ersche et al., 2012b). There were also significant differences between unaffected siblings and controls in performance on executive function tasks and trait measures of global impulsivity (BIS total score) and sensitivity to negative affect, related to the trait of negative urgency; however, in all these instances there were also differences between affected and non-affected siblings (Ersche et al., 2012b). A recent co-twin study has also attempted to identify which trait and cognitive impulsivity markers are implicated in vulnerability to SUD. They found that, although a trait measure of conscientiousness was not altered in non-affected twins, a general measure of disinhibition (i.e. reckless behaviour or rash impulsivity) and an electrophysiological marker of inhibitory control (the P300 potential evoked by an attentional interference task) were significantly altered in both affected and unaffected twins (Joyner et al., 2020).

It is expected that these trait and neurocognitive alterations reflect back on structural and functional neuroimaging measures. In Ersche and colleagues' cohort, structural neuroimaging measures uncovered brain regions with overlapping abnormalities in both stimulant users and their siblings. Relative to healthy controls, both affected and unaffected siblings exhibited larger volumes of the amygdala and putamen, and lower volumes of the insula, postcentral and superior temporal gyri (Ersche et al., 2012a). More recently, the same group has identified differential patterns of network communication across the brain in affected and non-affected siblings. Specifically, pairs sharing family risk showed weakened functional connectivity between the medial prefrontal cortex, the orbitofrontal cortex, the rostral anterior cingulate cortex and the caudate (Ersche et al., 2020). This network has been previously associated with reward valuation and goal-orientated decision-making, and weakened connections may reflect dysfunctions in this mechanism, which is pivotal to make adaptive decisions in "hot" scenarios, whereby the anticipated reward and inherent risks of different options are uncertain and complex.

A second approach that can speak to the role of impulsivity as an endophenotype for SUD are studies in offspring of people affected by the disorder, but which have no (or minimal) exposure to substance use themselves. In this case, impulsivity may be ascribed to shared vulnerability (i.e., endophenotype) and/or contextual factors such as parental influences. A recent study investigated which aspects of trait impulsivity, measured with the BIS, were elevated in children (aged 10–12) with a family history of SUD, compared to those without family history (Acheson et al., 2016). Findings showed that at-risk children had significantly elevated levels of non-planning and attention impulsivity compared with those without family risk. Within the same study, a longitudinal follow up into mid adolescence (15 years old) indicated that these impulsivity levels were quite consistent across time, suggesting direct endophenotype effects. Within the same cohort, another study examining the cognitive processes of motor impulsivity or impulsive action (go/no-go task) and impulsive choice (delay discounting) by age 10–12, children with family history showed deficits in both aspects; however, impulsive action deficits were no longer observed at mid adolescence (Dougherty et al., 2015). Adults with family history of SUD have also shown elevated non-planning impulsivity traits and cognitive planning deficits, which are similar to those observed in socio-demographically matched participants with SUD (Khemiri et al., 2020a). Heightened non-planning impulsivity traits but normative stop signal task performance in adults with family history of SUD was further confirmed in an extended international two-site study (Khemiri et al., 2020b).

In neuro-pharmacological/imaging research, young adults with family history of SUD but minimal levels of use (i.e. at risk), relative to controls with no family history, exhibited increased reward-evoked dopamine response measured by PET/MRI during the MID task (Weiland et al., 2017). Young adults at risk of SUD by virtue of trait impulsivity-novelty seeking showed similar increases in dopamine function in extra-striatal areas including the basolateral amygdala, the insula and the medial prefrontal cortex, middle and superior frontal gyri (Jaworska et al., 2020). Furthermore, 9-10-year-old children with family history of SUD enrolled in the multi-site ABCD study (n = 951), compared to the rest of the cohort (total n = 6898), displayed increased activation in cortical areas implicated in executive control during successful stop signal response inhibition (superior parietal cortex and paracentral gyri) (Lees et al., 2020). In a sub-analysis focused on maternal history of SUD, family history was associated with increased cerebellar activation during unsuccessful inhibition, a key process in cognitive-motor impulsivity, specifically among females (Lees et al., 2020). The latter finding suggest the need to further explore gender differences in the context of impulsivity vulnerabilities.

Altogether, the findings of endophenotype studies suggest that non-planning impulsivity and emotional sensitivity (akin to negative urgency) traits confer vulnerability to SUD, whereas increases in sensation seeking are a consequence of SUD. The evidence is less consistent for cognitive processes. Impulsive action deficits measured with stop signal performance and related brain activation have been found in some studies (Ersche et al., 2012b; Lees et al., 2020) but not others (Dougherty et al., 2015; Khemiri et al., 2020a,b). Future studies on impulsive action could benefit from applying recently proposed consensus recommendations to harmonize the design features and analysis of stop signal performance (Verbruggen et al., 2019). There is not sufficient evidence on impulsive choice, and other cognitive processes such as attentional interference and planning-related impulsivity may be associated with SUD vulnerability according to these studies. Increased dopamine turnover and weakened connectivity in the medial orbitofrontal striatal network were found to contribute to vulnerability for SUD.

4. Shared vulnerability between substance use and behavioural addictive disorders

In this section, we review studies that have compared trait features and cognitive performance between participants with SUD and those with gambling disorder (GD), a behavioural addiction that shares vulnerability factors with SUD (i.e. indicator of risk), and healthy controls (i.e. no risk). Although GD often co-occurs with alcohol and substance use, these studies recruited gamblers without history of moderate/severe alcohol use or drug use. In this context, similarities between participants with SUD and GD can be interpreted as indicative of shared vulnerabilities.

In a behavioural study examining trait facets (measured with the UPPS-P scale) and cognitive indices of impulsivity (Stroop test), findings showed that positive urgency (acting impulsively under positive affect) was elevated in both participants with stimulant-related SUD and those with GD, relative to controls (Albein-Urios et al., 2012). Conversely, negative urgency and cognitive indices of other executive functions (i.e. working memory) were altered specifically among participants with SUD in the same cohort. A subsequent functional neuroimaging study in the same cohort examined brain activation differences between stimulant-related SUD and GD and healthy controls during a reversal learning task, which captures elements of reward-related action selection and behaviour control, key aspects in cognitive impulsivity (Verdejo-Garcia et al., 2015). Both participants with SUD and GD showed reduced right ventrolateral prefrontal cortex (VLPFC) activation during reversal shifts. In contrast, only participants with SUD showed reduced dorsolateral prefrontal cortex activation and behavioural deficits in the task. These findings were further qualified in a recent structural MRI study that showed that both participants with stimulant-related SUD and GD exhibit structural abnormalities (i.e. larger brain volumes) in the right inferior frontal gyrus, i.e. adjacent to the right VLPFC region observed in the fMRI study (Irizar et al., 2020).

Three other neuroimaging studies have compared participants with SUD (tobacco and cocaine) and GD in task-evoked fMRI studies using motor impulsivity measures such as the stop signal task and choice impulsivity gambling-related reward tasks involving probabilistic wins and losses. In the fMRI-stop signal study, participants with tobacco SUD and GD showed overlapping reductions in the activation of the dorsomedial prefrontal/anterior cingulate cortex during both correct and failed inhibition trials (de Ruiter et al., 2012). In the gambling tasks studies, participants with cocaine SUD and GD showed similar increases in frontostriatal activation (middle and superior frontal, anterior cingulate, insula, striatum and midbrain) during reward anticipation (an experimental condition similar to that measured by the MID); however, participants with GD showed specific loss-related reductions of activation in the medial prefrontal cortex and anterior cingulate (Worhunsky et al., 2014, 2017). These studies suggest that abnormal ups and downs in the activation of medial prefrontal regions preceding reward receipt (up) and control (down) are commonly associated with the vulnerability to SUD and behavioural addictive disorders.

Altogether, the findings of clinical reference group-based comparison studies suggest that positive urgency impulsivity traits, impulsive action deficits and increased medial prefrontal/anterior cingulate and striatal response during reward valuation are shared vulnerability aspects of substance use and behavioural addictive disorders, reflecting SUD vulnerability. These insights from clinical populations should be validated as prognostic markers in longitudinal and endophenotype studies of at-risk cohorts.

5. New insights on the relationship between impulsivity-related vulnerabilities and SUD

The studies reviewed here corroborate the notion that impulsivity is a strong vulnerability factor for the development of SUD (Miller et al., 2018; Vassileva and Conrod, 2019; Verdejo-Garcia et al., 2008). The relationship between impulsivity and SUD vulnerability stands after considering other individual differences and contextual risk factors (Castellanos-Ryan and Conrod, 2020; Mackinnon et al., 2014). Emerging evidence suggests that the impulsive facets of non-planning impulsivity and emotion-driven impulsivity (i.e., positive and negative urgency), are particularly linked to the risk of developing substance use related problems (Albein-Urios et al., 2012; Ersche et al., 2010; Mackinnon et al., 2014). There is no indication that these factors predispose to using different substances, as they have been linked to different classes of drugs such as alcohol and stimulants (Acheson et al., 2016; Albein-Urios et al., 2012; Ersche et al., 2010; Pedersen et al., 2016). The evidence for specific aspects of cognitive functioning is comparatively more tenuous, and findings across different studies are mixed (Ersche et al., 2012b; Gustavson et al., 2017; Jones et al., 2020; Romer Thomsen et al., 2018). This could be due to methodological limitations in existing cognitive tasks of motor, choice and reflection impulsivity, such as limited reliability and inability to capture sufficient variability (Enkavi et al., 2019), and disparity in assessment methods, as there is substantial variation in the selection of cognitive tasks across different studies (Verdejo-Garcia et al., 2020). We discuss the findings for impulsive action and choice separately in the following paragraphs.

Impulsive action has been linked with vulnerability in (some) longitudinal, endophenotype and clinical reference studies (Albein-Urios et al., 2012; de Ruiter et al., 2012; Ersche et al., 2012b; Romer Thomsen et al., 2018). In functional neuroimaging studies, response inhibition related activity in right lateral prefrontal regions has also been associated with SUD vulnerability across different methodological approaches (Morein-Zamir et al., 2013; Romer Thomsen et al., 2018; Verdejo-Garcia et al., 2015). However, well-established behavioural indices of this construct, such as the stop signal reaction time, go/no-go commission errors or Stroop inhibition scores do not longitudinally predict development of SUD (Gustavson et al., 2017; Jones et al., 2020; Romer Thomsen et al., 2018). Furthermore, stop signal performance is well-preserved among people at risk of SUD by virtue of family history (Khemiri et al., 2020a). It is possible that neuroimaging indices are more sensitive to individual differences in response inhibition, and in particular to subtle variations in the latent characteristics that longitudinally predict SUD risk (Whelan et al., 2014).

The research on the relationship between impulsive choice and SUD vulnerability has similarly yielded miscellaneous findings. Delay discounting is longitudinally associated with tobacco smoking (Audrain-McGovern et al., 2009), but a methodologically strong study using a comprehensive test battery of discounting measured did not find a longitudinal association with alcohol related problems (Bernhardt et al., 2017). These disparate findings could reflect a specific link between impulsive choice and risk of tobacco use, but also highlight the need to increase consistency in the selection of assessment tools to enable comparability across studies. The timing of assessments is another key factor, as delay discounting measures collected earlier in life may be better indicators of future risk, and the availability of multiple measures across time can enable detailed analyses of developmental trajectories (Martinez-Loredo et al., 2018). Furthermore, the relationship between delay discounting and vulnerability to SUD seems to rely on its interaction with other executive functions such as working memory (Khurana et al., 2013, 2017). Delay discounting measures have not been consistently included in endophenotype studies, and suggest disorder-specific (rather than shared vulnerability) mechanisms in clinical reference group comparison studies (Albein-Urios et al., 2014). There is an alternative possibility that impulsive delay discounting tendencies reflect a general predisposition towards externalising psychopathology, rather than to SUD/addictive disorders specifically (Amlung et al., 2019). Delay discounting can also be a mediator of the relationship between early developmental trauma and subsequent SUD risk (Oshri et al., 2018).

In contrast with the mixed findings on behavioural measures of impulsive choice, functional neuroimaging measures of reward-related responsivity, such as reward anticipation during the Monetary Incentive Delay task (MID), are strong and consistent predictors of vulnerability to alcohol and drug related problems (Buchel et al., 2017; Whelan et al., 2014). Furthermore, MID-evoked indices of dopaminergic function measured with MR-PET (i.e., frontostriatal D2-type receptors) are altered among young adults at risk of SUD via trait predispositions and family history (Jaworska et al., 2020; Weiland et al., 2017). The MID task engages reward valuation processes similar to delay discounting, but lacks the key action selection aspect of impulsive choice. Thus, it is possible that MID findings reflect neurocognitive aspects of impulsivity that are not covered by current theoretical frameworks and behavioural measures. Recently, we have proposed and validated an alternative cognitive model of impulsivity and a comprehensive test battery to measure reward-related action selection (Verdejo-Garcia et al., 2020). Future studies should evaluate the relevance of this framework and measures for SUD vulnerability.

Despite the large body of literature and impressive advances in methodological approaches, power, and integration of novel technologies (e.g., neuroimaging, machine learning), there is still a dearth of studies that include comprehensive assessments of the different aspects of impulsivity (i.e., trait, motor, choice). Since there is now emerging agreement that these different aspects are independent contributors to impulsivity-related vulnerabilities, future studies should incorporate well-validated measures for each of them. We should also explore if there are other aspects of impulsivity that are not covered in current frameworks; for example, the anticipatory reward-related reactions captured by the MID task cannot be easily classified as either action or choice impulsivity. The reward-related aspects of impulsivity, and their relationship with both rapid-response actions and choices that seek immediate gratification is not yet well covered within existing frameworks and taxonomies (Verdejo-Garcia et al., 2020). In addition to "covering all bases", greater consistency across different measures of the same construct would be desirable to increase confidence and comparability within this field of research. That is, future studies should include several measures of the same construct (e.g., stop signal and go/no-go for impulsive action), and consider only convergent evidence from both measures as indicative of reliable associations with SUD-related measures. Furthermore, there is a need to harmonize the outcome measures employed to indicate SUD and their proxies. Currently, there are multiple definitions of "problematic" alcohol and drug use and thus very diverse outcomes across different studies (e.g., binge drinking, exposure to illicit drugs, escalation of drug use, SUD diagnoses). It is therefore difficult to generalise vulnerability factors observed in a particular cohort and set of outcomes to other samples and the population of reference. Furthermore, although here we focused on vulnerability, impulsivity is also associated with later stages of SUD, such as loss of control over drug use and difficulty to refrain from use (Stevens et al., 2014) and there is a similar need for predictive markers in this space. Recently, we have launched several initiatives to promote international collaborations in this area, which could be used to reach consensus on the assessment of impulsivity and outcome measures in SUD research (Verdejo-Garcia et al., 2019).

6. Future implications

Notwithstanding the limitations discussed in the previous section, the strong link between trait and cognitive impulsivity-related dispositions, which can be measured before onset of substance use, and later development of SUD has important implications for our surveillance systems and early intervention efforts. Pragmatic (i.e., short, easy to administer) measures of trait and cognitive impulsivity can be incorporated into national surveys, population-based studies and educational settings that have leverage to identify and buffer the relationship between dysfunctional impulsive characteristics and SUD risk. Future studies should exploit these early assessment opportunities to explore the relationship between trait and cognitive impulsivity and novel patterns of drug use, such as new tobacco related products and vaping, and recently legalised cannabis and cannabinoid products. Gender, and gender by adolescent development effects have been highlighted in recent studies (Lees et al., 2020), and deserve further investigation as little is known about trait or cognitive impulsivity-related SUD vulnerabilities in girls compared to boys. The short- and long-term impacts of the COVID-19 pandemic lockdowns and social isolation on trait and cognitive impulsive predispositions, which are an intrinsic feature of adolescent exploration and autonomy-seeking (Sparks, 2019), are also worth exploring. Another booming area in impulsivity research is the growing impact of screen exposure and related sleep alterations on impulsive behaviour and SUD risk during adolescence (Logan et al., 2018). Future research should also delve into the existing and future possibilities to apply universal and/or targeted prevention approaches that tackle dysfunctional impulsive traits and cognitive processes (Castellanos-Ryan and Conrod, 2020; Conrod et al., 2013). This area is still in need of evidence-based approaches that can be implemented in primary settings (US Preventive Services Task Force, 2020). These approaches could take the form of strengths-based programs that attempt to train different aspects of cognitive impulse control via neurocognitive training and remediation, and/or psychosocial interventions to build resilience against SUD, or targeted preventative programs designed to buffer the impact of at-risk predispositions. Given the strong and consistent link between impulsivity and SUD vulnerability, future research must aspire to identify, validate and implement strategies to curb this relationship and build enduring resilience against substance-related problems.

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